

```
import pandas as pd

from sklearn.datasets import load_iris

# Load dataset
iris = load_iris()

X = iris.data
y = iris.target

print("Dataset loaded successfully!")
print("Features:", iris.feature_names)
print("Target names:", iris.target_names)
```

```
Dataset loaded successfully!
Features: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal
Target names: ['setosa' 'versicolor' 'virginica']
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("Training data:", len(X_train))
print("Testing data:", len(X_test))
```

```
Training data: 120
Testing data: 30
```

```
from sklearn.tree import DecisionTreeClassifier

# Create model
model = DecisionTreeClassifier(random_state=42)

# Train model
model.fit(X_train, y_train)

print("Model trained successfully!")
```

```
Model trained successfully!
```

```
from sklearn.metrics import accuracy_score

# Predict
predictions = model.predict(X_test)
```

```
# Accuracy
accuracy = accuracy_score(y_test, predictions)

print("Accuracy:", accuracy * 100, "%")
```

Accuracy: 100.0 %

```
# Sepal length, Sepal width,
# Petal length, Petal width

new_flower = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(new_flower)

print("Predicted Flower Species:",
      iris.target_names[prediction[0]])
```

Predicted Flower Species: setosa

```
df = pd.DataFrame(
    iris.data,
    columns=iris.feature_names
)

df["species"] = [
    iris.target_names[i] for i in iris.target
]

print(df.head(10))
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	
5	5.4	3.9	1.7	0.4	
6	4.6	3.4	1.4	0.3	
7	5.0	3.4	1.5	0.2	
8	4.4	2.9	1.4	0.2	
9	4.9	3.1	1.5	0.1	

	species
0	setosa
1	setosa
2	setosa
3	setosa
4	setosa
5	setosa
6	setosa
7	setosa

```
8 setosa
```

```
9 setosa
```